Final Report: NUCE 431W

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Project Outline

With the intention for the spherical tokamak advanced reactor (STAR) to generate energy; It uses a D-T reaction to generate high-energy neutrons. However, there are multiple issues associated with the use of tritium in this reaction. The first is that tritium has a relatively short half-life of 2 years, which makes it difficult to accumulate and store for this reaction. Second, the energy emitted in this reaction is mostly transferred to the emitted neutron, leading to extremely high energy neutrons, 14 MeV. To resolve these issues, lithium-6 is used with a neutron source to produce both tritium and a small number of neutrons. This occurs within the blanket, which contains the lithium-6 to breed tritium. Once the D-T reaction occurs, the kinetic energy of the 14 MeV neutrons must be converted to a more useful energy form. The current moderator of choice, within the blanket, is lead. The primary focus of our work is optimizing the blanket's configuration to promote tritium production for this D-T reaction. The secondary focus is the optimization of the neutron shield to protect other components and personnel.

Tritium, being an extremely expensive and scare natural resource, is needed for successful fusion within the STAR reactor. A goal of the STAR reactor is to produce its own Tritium within the blanket region of the reactor. For this production, Lithium is used. There exists a large probability for Lithium to produce Tritium when interacting with a neutron. Since the probability of interaction is quite low for high energy neutrons like the ones given off in a D-T fusion reaction, neutron multipliers and moderators are used within the breeding blanket to maximize Tritium production.

Consequentially, the Lithium is not conserved when it is turned into Tritium. For this reason the breeding mixture needs to be cycled out periodically to replace lost Lithium content and extract the Tritium produced. For an efficient closed-loop extraction system, the breeder mixture shall be liquid to facilitate this process.

Objectives

Our group was assigned with two main tasks; Maximize Tritium produced within the breeder layer and minimize neutron flux to critical components. To accomplish this, two aspects of the STAR reactor were changed,

- **Radial orientation of Materials**
- **Materials themselves**

Changing the orientation of the materials requires changing the materials associated to each part. For example, the following two configurations were tested with materials at different radial positions,

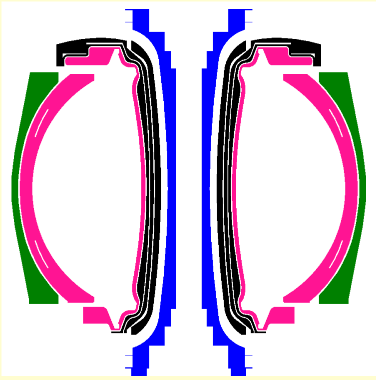

Figure 1: Config. 1

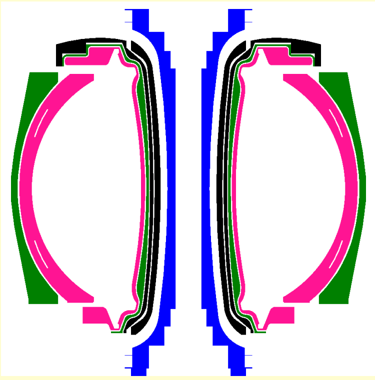

Figure 2: Config. 5

In these two figures; Green is multiplier, Pink is breeder, Black is shielding, and Blue is steel. The difference between the two is the inner shielding configuration. The shielding closest to the breeder on the radial inside was changed to multiplier to see the affect of neutron multiplication when placed behind the breeder.

Not only this, but the materials themselves were changed within the shield, breeder, and multiplier. Many different known breeders were tested, each with different levels of Lithium enrichment. As a side effect of our testing, validation and verification of PPPL's numbers was done. Since they were using a different neutronics code to calculate the TBR's for different materials, our work also served as a sanity check to ensure their results were realistic.

Results

Overall, we have accomplished the goal of calculating and testing the TBR for various materials and geometric configurations. We have tabulated the results based on material used. The results of our work are as follows. (Each table is paired with a configuration),

Configuration 1: Multiplier outside plasma

Figure 3: Configuration 1 TBR and Data

	CONFIGURATION 1: config1_scaled.h5m										
Shield	Mult.	Density	Breeder	At. %	Density	Enriched?	Enrichmen	Enrichment	Particles	TBR	Margin
B4C	Pb	10.5	PbLi	83/17	9.5	N	N/A	nat.	1.00E+04	0.560734	0.00183844
B4C	Pb	10.5	PbLi	83/17	9.5	Y	Li-6	30.00%	1.00E+04	0.944533	0.00327229
B4C	Pb	10.5	PbLi	83/17	9.5	Y	Li-6	60.00%	1.00E+04	1.14721	0.00404799
B4C	Pb	10.5	PbLi	83/17	9.5	Y	Li-6	90.00%	1.00E+04	1.25379	0.00369134
B4C	Pb	10.5	Li	N/A	0.48	N	N/A	nat.	1.00E+04	1.15823	0.00242229
B4C	Pb	10.5	Li	N/A	0.48	Y	Li-6	30.00%	1.00E+04	1.20372	0.00363703
B4C	Pb	10.5	Li	N/A	0.48	Y	Li-6	60.00%	1.00E+04	1.08875	0.00298815
B4C	Pb	10.5	Li	N/A	0.48	Y	Li-6	90.00%	1.00E+04	0.940676	0.00188553
B4C	Pb	10.5	Li2O	nat.	2.013	N	N/A	nat.	1.00E+04	1.23702	0.00257002
B4C	Pb	10.5	Li2O	N/A	2.013	Y	Li-6	30.00%	1.00E+04	1.18521	0.00324064
B4C	Pb	10.5	Li2O	N/A	2.013	Y	Li-6	60.00%	1.00E+04	1.06309	0.00289393
B4C	Pb	10.5	Li2O	N/A	2.013	Y	Li-6	90.00%	1.00E+04	0.937327	0.00286696
H2O	Pb	10.5	PbLi	83/17	9.5	N	N/A	nat.	1.00E+04	0.75226	0.00222745
Cd	Pb	10.5	PbLi	83/17	9.5	N	N/A	nat.	1.00E+04	0.623938	0.00291449
Cd	Cd	8.65	PbLi	83/17	9.5	N	N/A	nat.	1.00E+04	0.5265	0.00243548

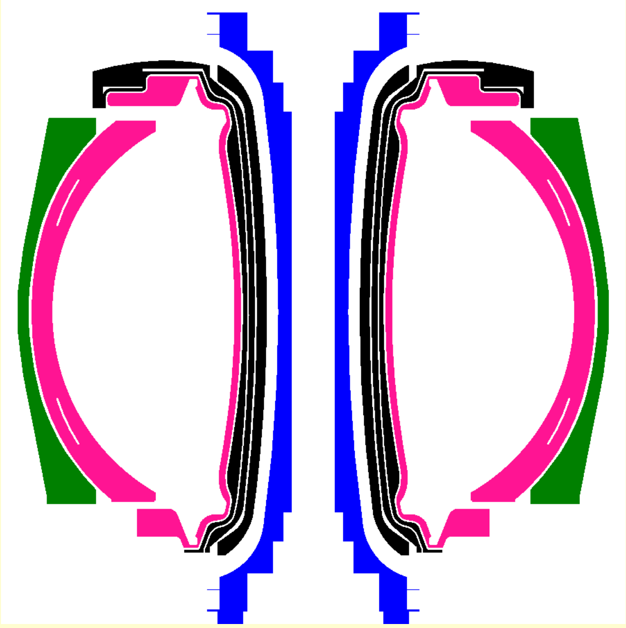

Figure 4: Y-Z Cross Section of Configuration 1

Configuration 2: Shield-Swapped with Multiplier

Figure 5: Configuration 2 TBR and Data

CONFIGURATION 2: config2_scaled.h5m

Mult.	Density	Breeder	% Pb/Li	Density	Enriched?	Enrichmer	Enrichmer	Particles	TBR	Margin
Pb	10.5	PbLi	83/17	9.5	N	N/A	nat.	1.00E+04	0.563937	0.00194808
Pb	10.5	PbLi	83/17	9.5	Y	Li-6	30.00%	1.00E+04	0.949603	0.00326127
Pb	10.5	PbLi	83/17	9.5	Y	Li-6	60.00%	1.00E+04	1.15204	0.00428893
Pb	10.5	PbLi	83/17	9.5	Y	Li-6	90.00%	1.00E+04	1.25952	0.00383138
Pb	10.5	Li	83/17	0.48	N	N/A	nat.	1.00E+04	1.1638	0.00204491
Pb	10.5	Li	83/17	0.48	Y	Li-6	30.00%	1.00E+04	1.21047	0.00356322
Pb	10.5	Li	83/17	0.48	Y	Li-6	60.00%	1.00E+04	1.09476	0.00310595
Pb	10.5	Li	83/17	0.48	Y	Li-6	90.00%	1.00E+04	0.946312	0.0019997

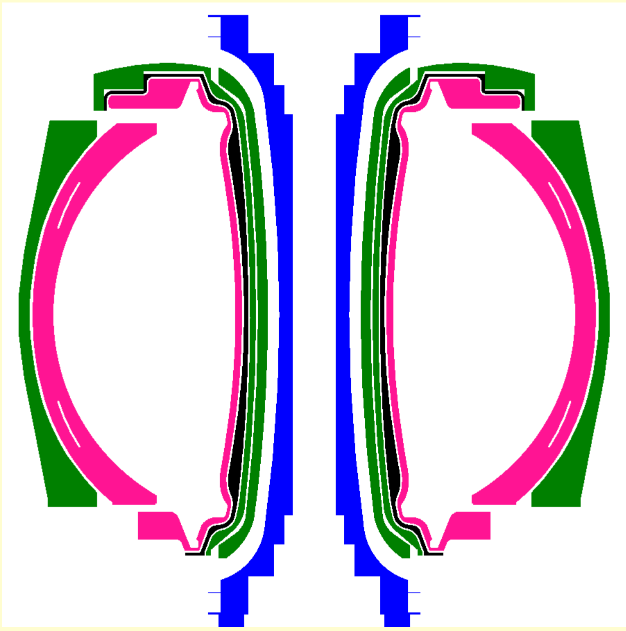

Figure 6: Y-Z Cross Section of Configuration 2

Configuration 3: Breeder-Swapped Radially with Multiplier

Figure 7: Configuration 3 TBR and Data

CONFIGURATION 3: config3_scaled.h5m										
Mult.	Density	Breeder	% Pb/Li	Density	Enriched?	Enrichmer	Enrichmer	Particles	TBR	Margin
Pb	10.5	PbLi	83/17	9.5	N	N/A	nat.	1.00E+04	0.366273	0.00161196
Pb	10.5	PbLi	83/17	9.5	Y	Li-6	30.00%	1.00E+04	0.627508	0.00197351
Pb	10.5	PbLi	83/17	9.5	Y	Li-6	60.00%	1.00E+04	0.793246	0.00288903
Pb	10.5	PbLi	83/17	9.5	Y	Li-6	90.00%	1.00E+04	0.900544	0.00270838
Pb	10.5	Li	83/17	0.48	N	N/A	nat.	1.00E+04	0.594196	0.00182034
Pb	10.5	Li	83/17	0.48	Y	Li-6	30.00%	1.00E+04	0.991192	0.00256757
Pb	10.5	Li	83/17	0.48	Y	Li-6	60.00%	1.00E+04	1.17096	0.00290646
Pb	10.5	Li	83/17	0.48	Y	Li-6	90.00%	1.00E+04	1.25269	0.00253186

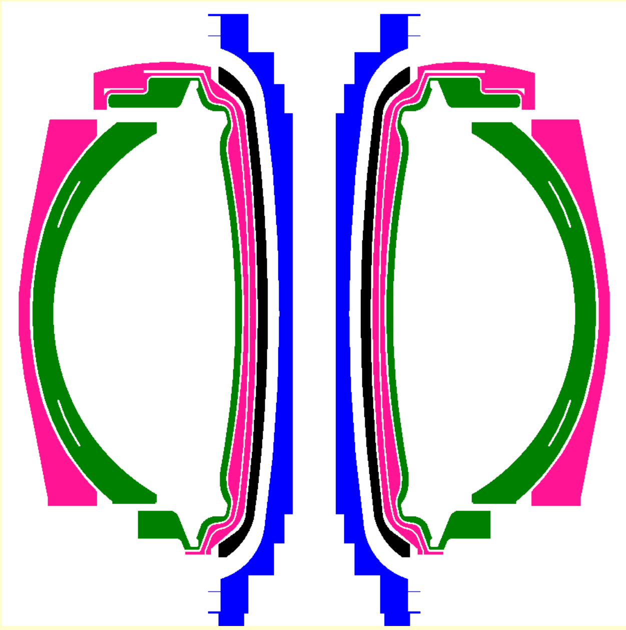

Figure 8: Y-Z Cross Section of Configuration 3

Configuration 5: Breeder-Swapped Radially with Multiplier

Figure 9: Configuration 5 TBR and Data

	CONFIGURATION 5: config5_scaled.h5m										
Shield	Mult.	Density	Breeder	At. %	Density	Enriched?	Enrichme	Enrichme	Particles	TBR	Margin
B4C	Pb	10.5	PbLi	83/17	9.5	N	N/A	nat.	1.00E+04	0.654377	0.00277739
B4C	Pb	10.5	PbLi	83/17	9.5	Y	Li-6	30.00%	1.00E+04	1.04405	0.00282089
B4C	Pb	10.5	PbLi	83/17	9.5	Y	Li-6	60.00%	1.00E+04	1.2414	0.00432882
B4C	Pb	10.5	PbLi	83/17	9.5	Y	Li-6	90.00%	1.00E+04	1.34405	0.00341228
B4C	Pb	10.5	Li	N/A	0.48	N	N/A	nat.	1.00E+04	1.23921	0.00322416
B4C	Pb	10.5	Li	N/A	0.48	Y	Li-6	30.00%	1.00E+04	1.27402	0.00295753
B4C	Pb	10.5	Li	N/A	0.48	Y	Li-6	60.00%	1.00E+04	1.15187	0.00230246
B4C	Pb	10.5	Li	N/A	0.48	Y	Li-6	90.00%	1.00E+04	0.998898	0.00212503
B4C	Pb	10.5	Li2O	N/A	2.013	N	N/A	nat.	1.00E+04	1.28241	0.00278581
B4C	Pb	10.5	Li2O	N/A	2.013	Y	Li-6	30.00%	1.00E+04	1.21513	0.00358379
B4C	Pb	10.5	Li2O	N/A	2.013	Y	Li-6	60.00%	1.00E+04	1.08547	0.00311344
B4C	Pb	10.5	Li2O	N/A	2.013	Y	Li-6	90.00%	1.00E+04	0.956268	0.0030675

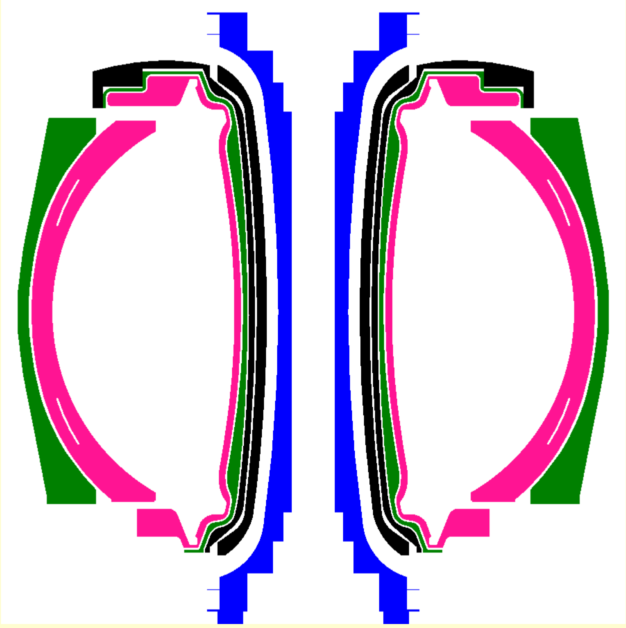

Figure 10: Y-Z Cross Section of Configuration 5

Original Configuration, Exotic Material Testing

Along with these different configurations, the original design given by PPPL was tested with more exotic breeding material. The results are tabulated below, with increasing amount of Lithium-6 enrichment,

Figure 11: Natural Li-6 Content

Particles	TBR (n,Xt)	Margin	density
Li4SiO4			
10K	0.893613	0.003069	2.39
Li8ZrO6			
10K	1.05849	0.0030058	2.58
Li2O			
10K	1.22625	0.0025112	2.013
LiAlO2			
10K	0.697136	0.0028363	2.62
Li5AlO4			
10K	0.98227	0.0038548	2.17
Li2ZrO3			
10K	0.933997	0.0028539	4.15
Li2TiO3			
10K	0.891498	0.0020768	3.43
(nat. Li)			

Figure 12: 30% Li-6 Content

Particles	TBR (n,Xt)	Margin	density
Li4SiO4			
10K	0.896543	0.0034513	2.39
Li8ZrO6			
10K	1.08168	0.00285	2.58
Li2O			
10K	1.1823	0.0031713	2.013
LiAlO2			
10K	0.747349	0.0028336	2.62
Li5AlO4			
10K	0.984343	0.0036581	2.17
Li2ZrO3			
10K	1.01877	0.0039677	4.15
Li2TiO3			
10K	0.929748	0.0015646	3.43
(30% Li-6)			

Figure 13: 60% Li-6 Content

Li4SiO4			
Particles	TBR (n,Xt)	Margin	density
10K	0.839592	0.002236	2.39
Li8ZrO6			
Particles	TBR (n,Xt)	Margin	density
10K	1.02502	0.0025633	2.58
Li2O			
Particles	TBR (n,Xt)	Margin	density
10K	1.06147	0.0029804	2.013
LiAlO2			
Particles	TBR (n,Xt)	Margin	density
10K	0.73861	0.0019521	2.62
Li5AlO4			
Particles	TBR (n,Xt)	Margin	density
10K	0.91833	0.0034232	2.17
Li2ZrO3			
Particles	TBR (n,Xt)	Margin	density
10K	1.01752	0.0043505	4.15
Li2TiO3			
Particles	TBR (n,Xt)	Margin	density
10K	0.910904	0.00245	3.43
(60% Li-6)			

Figure 14: 90% Li-6 Content

Li4SiO4			
Particles	TBR (n,Xt)	Margin	density
10K	0.775256	0.0028558	2.39
Li8ZrO6			
Particles	TBR (n,Xt)	Margin	density
10K	0.95221	0.0030839	2.58
Li2O			
Particles	TBR (n,Xt)	Margin	density
10K	0.936234	0.0028167	2.013
LiAlO2			
Particles	TBR (n,Xt)	Margin	density
10K	0.716108	0.0022283	2.62
Li5AlO4			
Particles	TBR (n,Xt)	Margin	density
10K	0.839889	0.0023513	2.17
Li2ZrO3			
Particles	TBR (n,Xt)	Margin	density
10K	0.992885	0.0031823	4.15
Li2TiO3			
Particles	TBR (n,Xt)	Margin	density
10K	0.869415	0.0020949	3.43
(90% Li-6)			

Figure 15: FLiBe (nat. Li-6 Content)

FLiBe			
Particles	TBR (n,Xt)	Margin	density
10K	1.04452	0.0031095	2

Figure 16: FLiNaK (nat. Li-6 Content)

FLiNaK			
Particles	TBR (n,Xt)	Margin	density
10K	0.868961	0.003059	2
(nat. Li)			

Discussion

Overall, these results lined up well with what we were expecting to see. As seen above, some of these configurations were very promising, specifically configuration 5. As seen in Fig. 9, the TBR for Lithium Oxide (with natural concentration of Li-6) performed very well, with a TBR of ~ 1.28. This is very promising for this type of breeder. The caveat with Lithium Oxide is that it is a solid at the operating temperature of the reactor. This makes it hard / impossible to cycle through a closed loop, since it does not have the benefits of being a liquid. Through some surface level research, it was confirmed that Lithium Oxide can be a contamination byproduct of Lead-Lithium (PbLi) exposure to air and moisture. Since these are both promising breeders, allowed contamination of the breeder could be implemented to increase the amount of Lithium Oxide within the PbLi eutectic. To accompany this, the most performant results were observed when the multiplier **preceded** the breeder radially. This served as a direct moderator to any neutron entering the multiplier layer. It served as a scattering bed for slowing-down of neutrons, and as the name suggests, a multiplier. Neutrons released from (n,Xn) reactions in Lead are much lower energy compared to the energy of the neutron given off in the D-T reaction. These low energy neutrons have a **much** higher probability to produce Tritium with Lithium.

Not only this, but pure Lithium has a very promising TBR as well (~ 1.23), at natural Li-6 concentration. Interestingly, we observed a **decrease** in TBR when Li-6 concentration **increased**. This is extremely counterintuitive, since an increase in Li-6 should result in a massive increase in TBR. For a lot of these simulations, this is not the case. We have come to the conclusion that the reason for this is due to the absence of a multiplier and/or moderator. The scattering (inelastic and elastic) cross section for Li-6 is 100x smaller than Li-7. With less Li-7 to down-scatter neutrons into lower energies, more neutrons leak and leave the breeder. We hypothesized that this would result in a higher flux to the central magnet coil and increased flux to the outer shielding. This was confirmed when the mesh tally was checked outside of the reactor, we observed a much higher flux at this region.

Annular shielding was not changed in any of these configurations. Although tests showed that when this was changed, the TBR increased massively. This was offset by a high flux to the magnet coils. Optimally, the magnet coils should receive a flux as low as reasonably achievable. Any change in annular shielding (closest to magnet coil) resulted in a flux too high for the material.